

## Day 5 Of Communication System Class

Fourier transform representation of aperiodic signals , digital signals and its characteristics

|       |                                | Mathematical expressions                                                                                |
|-------|--------------------------------|---------------------------------------------------------------------------------------------------------|
| S.No. | Name of Property               |                                                                                                         |
| 1.    | Linearity of superposition     | $[a_1 x_1(t) + a_2 x_2(t)] \xrightarrow{F} [a_1 X_1(f) + a_2 X_2(f)]$ , $a_1$ and $a_2$ constants,      |
| 2.    | Time scaling                   | $x(\alpha t) \xrightarrow{F} \frac{1}{ \alpha } X\left(\frac{f}{\alpha}\right)$ , $\alpha$ is constant. |
| 3.    | Duality or symmetry            | If $x(t) \xrightarrow{F} X(f)$ then, $X(t) \xrightarrow{F} x(-f)$                                       |
| 4.    | Time shifting                  | $x(t - t_d) \xrightarrow{F} e^{-j2\pi f t_d} X(f)$                                                      |
| 5.    | Area under $x(t)$              | $\int_{-\infty}^{\infty} x(t) dt = X(0)$                                                                |
| 6.    | Area under $X(f)$              | $\int_{-\infty}^{\infty} X(f) df = x(0)$                                                                |
| 7.    | Frequency shifting             | $e^{j2\pi f_c t} x(t) \xrightarrow{F} X(f - f_c)$                                                       |
| 8.    | Differentiation in time domain | $\frac{d}{dt} x(t) \xrightarrow{F} j2\pi f X(f)$                                                        |

|     |                                |                                                                                                |
|-----|--------------------------------|------------------------------------------------------------------------------------------------|
| 5.  | Area under $x(t)$              | $\int_{-\infty}^{\infty} x(t) dt = X(0)$                                                       |
| 6.  | Area under $X(f)$              | $\int_{-\infty}^{\infty} X(f) df = x(0)$                                                       |
| 7.  | Frequency shifting             | $e^{j2\pi f_c t} x(t) \xrightarrow{F} X(f - f_c)$                                              |
| 8.  | Differentiation in time domain | $\frac{d}{dt} x(t) \xrightarrow{F} j2\pi f X(f)$                                               |
| 9.  | Integration in time domain     | $\int_{-\infty}^{\infty} x(\lambda) d\lambda \xrightarrow{F} \frac{1}{j2\pi f} X(f)$           |
| 10. | Conjugate functions            | If $x(t) \xrightarrow{F} X(f)$ , then $x^*(t) \xrightarrow{F} X^*(-f)$                         |
|     | Multiplication in time domain  | $x_1(t) x_2(t) \xrightarrow{F} \int_{-\infty}^{\infty} X_1(\lambda) X_2(f - \lambda) d\lambda$ |
|     | Convolution in time domain     | $x_1(t) * x_2(t) \xrightarrow{F} X_1(f) X_2(f)$                                                |

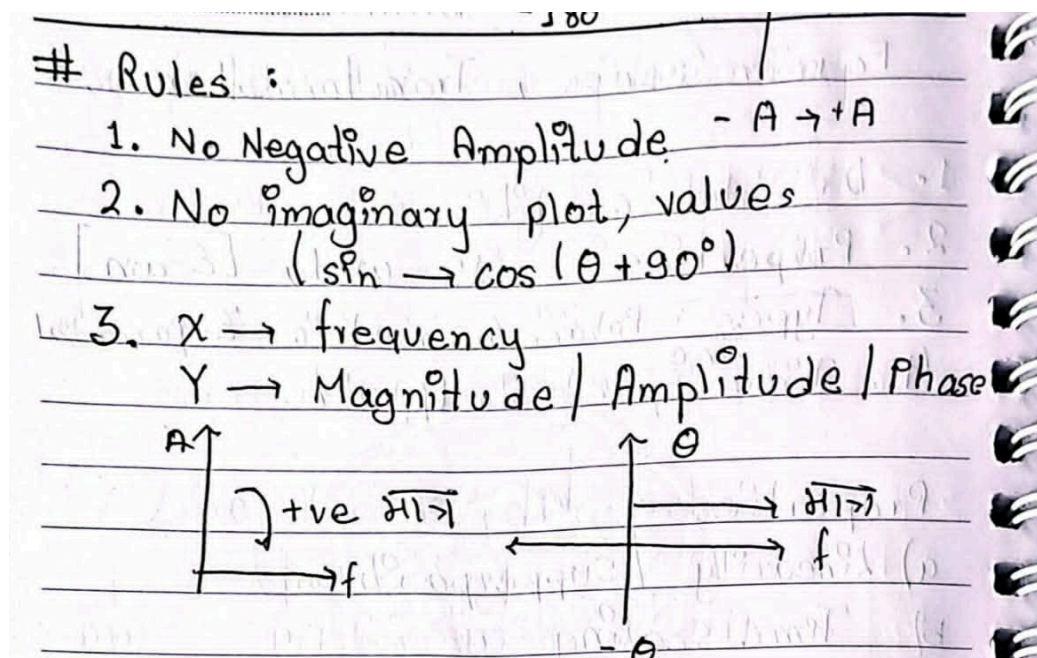
**For the given modulated signal draw line spectrums**

## Line Spectra

- The *amplitude spectrum* of  $x(t)$  is defined as the plot of the magnitudes  $|c_k|$  versus  $\omega$
- The *phase spectrum* of  $x(t)$  is defined as the plot of the angles  $\angle c_k = \arg(c_k)$  versus  $\omega$
- This results in *line spectra* which are defined for both positive and negative frequencies
- Notice: for  $k = 1, 2, \dots$

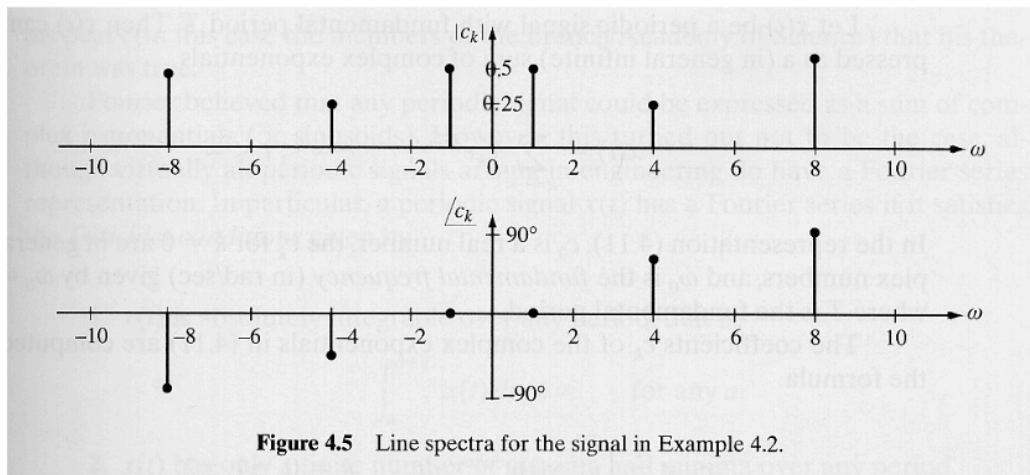
$$|c_k| = |c_{-k}| \quad \angle c_k = -\angle c_{-k}$$

$$\arg(c_k) = -\arg(c_{-k})$$



## Example: Line Spectra

$$x(t) = \cos(t) + 0.5 \cos(4t + \pi/3) + \cos(8t + \pi/2)$$



**Figure 4.5** Line spectra for the signal in Example 4.2.